Ailun (Allen) Shen

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RESEARCH INTERESTS

Efficient inference and parallel decoding in large language models; the dynamics by which learned capabilities are acquired, retained, and lost under fine-tuning.

RESEARCH EXPERIENCE

Researcher, Algoverse AI Research *June 2026 – Present*

- Investigating whether the parallel-decoding speedup of Consistency LLMs and Jacobi Forcing survives downstream supervised fine-tuning, measured by retention of tokens per forward pass.
- Reproducing published Jacobi-decoding baselines and re-measuring throughput after aggressive task-specific fine-tuning to quantify capability loss.
- Analyzing stationary tokens by linguistic category to localize where acceleration degrades, and developing consistency-preserving fine-tuning objectives to retain it.

Undergraduate Research Intern, University of Washington *Sept 2025 – Present*

Advisor: Prof. Shih-Chieh Hsu

- Integrating a Neural Simulation-Based Inference (NSBI) head into EveNet, a foundation model for particle-collision data, to enable likelihood-free parameter estimation in high-energy physics.
- Implementing and validating the inference head against existing analysis baselines; coordinating code review with the group's graduate researchers.

Independent Researcher — Gradient Descent Emerges via Natural Selection *July 2024 – Aug 2025*

- Designed a genetic algorithm testing whether evolutionary processes can produce gradient approximators sufficient for neural-network learning. Sole-author publication at IEEE CEC 2025.

Independent Researcher — Space-Time Curvature-Based Data Clustering *Sept 2023 – June 2024*

- Reformulated clustering using principles from general relativity and introduced the Curvature Density score, outperforming k-means and hierarchical baselines. First-author publication at IEEE AMLDS 2025.

PUBLICATIONS

1. A. Shen. "Gradient Descent Emerges via Natural Selection," *IEEE Congress on Evolutionary Computation (CEC)*, June 2025. (Sole author)
2. A. Shen et al. "Space-Time Curvature-Based Clustering," *IEEE AMLDS*, July 2025. (First author)
3. A. Shen et al. "Pervasive Seizure Detection via One-Class EEG Modeling with Visual Insights," *IEEE SITA*, October 2025. (Second author)

EDUCATION

University of Washington, Paul G. Allen School of Computer Science & Engineering *Seattle, WA*

B.S. in Computer Science — entering Autumn 2026 with junior standing. Expected June 2029.

Prior coursework: Linear Algebra & Differential Equations; IB Mathematics: Analysis HL and Applications & Interpretations HL; IB Physics HL; AP Calculus BC; AP Computer Science A. Interlake High School, Bellevue WA, GPA 3.986/4.0.

HONORS & SELECTED ACTIVITIES

- Selected participant, NSF IAIFI PhD Summer School, MIT (2026).
- Accepted poster presenter, NSF IAIFI Summer Workshop, Harvard University (2025).
- Bronze Medal and Student Ambassador, USA AI Olympiad (USAAIO), hosted by MIT (2025); Round 1 High Honor Roll (2026).
- Founder and Vice President, Interlake AI Club — teach AI foundations and run USAAIO preparation for student members.
- Co-Founder, PingPongParkinson WA Chapter — table-tennis-therapy nonprofit serving patients with Parkinson's disease.

TECHNICAL SKILLS

Languages: Python, Java
ML / Deep Learning: PyTorch, Hugging Face (Transformers, Datasets, PEFT), NumPy, scikit-learn
Tools: Git, Linux